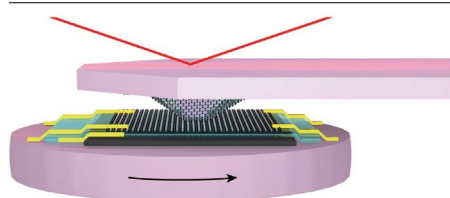


A new microscope puts a twist on 2D materials



Scanning probe microscopes have revolutionized our understanding of electronic phenomena by allowing us to visualize them at the atomic scale. However, they only probe the properties of materials at one point at a time. Now, writing in *Nature*, Shahal Ilani and colleagues present a conceptually new scanning probe microscope, which they call a quantum twisting microscope (QTM). The new instrument can probe the interference between electrons tunnelling into the sample through different paths and enables both momentum-resolved measurements and in situ tuning of the twist angle between 2D materials stacked on top of each other, systems that host fascinating physics.

The heart of the instrument is a 2D material placed on a rotating pyramid on an atomic force microscope cantilever. When brought into contact with the sample, the 2D material on the tip self-aligns with it, easily forming a clean interface. “This is an idea that my group has been thinking about for a long time, but 2 years ago we realized that now all the components were there, from methods to fabricate clean devices with 2D materials to proof-of-principle demonstrations of momentum-resolved measurements,” says Ilani. Moreover, the discovery of superconductivity in twisted bilayer graphene in 2018 created enormous interest around the study of twisted 2D materials, providing new momentum for the idea.

“We knew it could in principle work: it was still a crazy idea because millions of things could go wrong, but we did not overthink it and it worked perfectly, at first touch.”

The QTM can be used in two different ways. The first consists of investigating how the properties of the interface change depending on the angle between the two layers, which is exciting given the many intriguing phenomena observed in twisted 2D materials. The second option is to probe the energy bands of the sample, which is made possible by the momentum-resolved nature of the measurements. “The first direction enables measurements that might solve some of the mysteries in the field of twistrionics, whereas the second is more explorative: a momentum-conserving microscope is something that people wanted for a long time,” explains Ilani. “There is a good chance that very important discoveries will come: we basically have new eyes on quantum materials.”

The researchers report several proof-of-principle measurements. The first is the twist-dependent characterization of the interface between two graphene monolayers, realizing so far elusive in situ twistrionics measurements. The second is the momentum-resolved characterization of a twisted bilayer graphene sample. The third is the use of the QTM to apply pressures on the GPa scale at the interface while imaging changes in the energy bands.

Any combination of layered materials can be used for the tip and sample, including conductors, semiconductors and superconductors. Moreover, the setup is compatible with external magnetic fields and with cryogenic measurements, and the tip can be brought closer to the sample to apply pressure, taking the system into new

regimes of strong interactions. The availability of so many control knobs opens up many possibilities for experiments. “The possibilities are what we are most excited about,” comments John Birkbeck, lead author of the paper. “At the push of a button you can now scan the entire phase space and explore incredible physics: what previously would take a student their whole PhD can now be done in a day.”

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The next step will be to start measuring at cryogenic temperatures, entering the regime where the most interesting physics happens. The new microscope is also attracting the interest of several other groups who are starting to set up their own instruments. “The technique is versatile enough that people can take it in different directions,” concludes Ilani. “Theorists are also coming up with ideas, for example exploring what happens when you use two superconductors, which could help understand where superconductivity is coming from in momentum space, what is the symmetry of the order parameter and maybe create new types of superconductors. It’s a completely new front.”

Giulia Pacchioni

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